

Intraday Gold Price Forecasting Using Hybrid Machine Learning And Transformer Models Across Multiple Timeframes

Keywords: Financial Forecasting, Time Series Prediction, Hybrid LSTM, XGBoost, CatBoost, Informer, Walk-Forward Validation, Gold Price Prediction, Machine Learning, Deep Learning, Intraday Trading.

Abstract: This paper evaluates three machine learning architectures for intraday forecasting of gold prices (XAU/USD), using high-frequency bid and ask data collected between January 2024 and March 2025 at 5-minute, 15-minute, and 1-hour time-frames. The models compared include a Transformer-based Informer network and two hybrid architectures that combine Long Short-Term Memory (LSTM) networks with gradient-boosted tree regressors (XGBoost and CatBoost). All models were trained and assessed under a leakage-resistant walk-forward validation framework across multiple forecasting horizons. The empirical results show that the hybrid LSTM-based models consistently outperform the Informer model, particularly in shorter prediction horizons. These findings suggest that hybrid sequence-boosting architectures offer a more reliable forecasting approach for volatile intra-day commodity markets than the current generation of Transformer-based models.

1 INTRODUCTION

Gold holds a distinctive role in global financial markets as both a monetary asset and safe-haven hedge, with XAU/USD exhibiting strong liquidity and volatility during periods of macroeconomic uncertainty (Wen et al., 2022a). At high frequencies, gold price movements are influenced by order-flow imbalances, cross-asset linkages, and news-driven volatility bursts, producing regime shifts and microstructure noise that complicate short-term prediction (Liang et al., 2023).

Machine Learning (ML) and deep learning approaches have increasingly been adopted for financial time-series forecasting. Recent surveys highlight their growing impact across finance, energy, and cryptocurrency markets, with diverse architectures successfully applied to short-term prediction tasks (Kim et al., 2025; Sezer et al., 2020; Ferreira and Medeiros, 2021). Recurrent neural networks such as Long Short-Term Memory (LSTM) models are widely used due to their ability to capture temporal dependencies, while gradient-boosted decision tree models (e.g., XGBoost, CatBoost) remain highly competitive on structured features extracted from market data (Jabeur et al., 2024; Grinsztajn et al., 2022). More recently, Transformer-based architectures tailored for time series, such as Informer and PatchTST, have shown promise for long-range dependency modelling (Zhou et al., 2021; Nie et al., 2022). However, empirical evidence remains mixed: several studies report that Transformer models do not consistently outperform

classical or hybrid baselines, particularly in noisy, intraday financial contexts (Wen et al., 2022b).

Intraday forecasting of gold remains comparatively underexplored relative to equities and cryptocurrency markets, and many existing studies rely on random or single train-test splits, which risk temporal leakage and inflated performance (Wahyuddin et al., 2025). To address these limitations, this study applies a rigorous walk-forward, block-wise validation procedure to evaluate forecasting robustness under realistic market conditions. The dataset consists of bid and ask price data for XAU/USD from January 2024 to March 2025, aggregated into 5-minute, 15-minute, and 1-hour candlesticks (time-frames).

This research compares three state-of-the-art forecasting architectures: (1) a hybrid LSTM+XGBoost model, (2) a hybrid LSTM+CatBoost model, and (3) the Informer Transformer-based architecture. These models represent two contrasting methodological paradigms: recurrent sequence encoders enhanced by tree-based regression, and attention-driven sequence modelling. The results indicate that hybrid LSTM-boosting pipelines outperform the Informer across multiple prediction intervals, with LSTM+CatBoost achieving the strongest and most stable performance—particularly in short-interval intraday forecasts.

In summary, the contributions of this study are as follows: (1) A systematic, walk-forward evaluation of hybrid and Transformer-based architectures for intraday XAU/USD forecasting. (2) A comparison across multiple data granularities (5m,

15m, 1h) and forecasting intervals (5–60 minutes). (3) Empirical evidence demonstrating the superiority of hybrid LSTM–boosting models, especially LSTM+CatBoost, for volatile intraday commodity markets.

The rest of this paper is organised as follows: Section 2 reviews related work; Section 3 describes the data, preprocessing, and model architectures; Section 4 presents the experimental results; finally, the conclusions are drawn in Section 5.

2 RELATED WORK

Gold (XAU/USD) plays a dual role as a commodity and a monetary hedge, and its price dynamics are shaped by macroeconomic conditions, interest rate shifts and market sentiment (Baur and Lucey, 2010). Traditional econometric models such as ARIMA and GARCH have been widely used in precious metals forecasting. While these models were initially combined in early studies (e.g., Yaziz et al. (2013)) to better model volatility, more recent research continues to support this hybrid approach. For instance, Syrris and Shenai (2021) demonstrates that an ARIMA-GARCH model more accurately captures gold returns and volatility clustering, and Setyowibowo et al. (2022) shows that ARIMA-GARCH significantly outperforms pure ARIMA on daily gold price data.

With the growth of machine learning, recurrent neural networks — particularly LSTM networks — have become widely adopted for modelling sequential dependencies in financial time series (Nelson et al., 2017; Fischer and Krauss, 2018). LSTMs have demonstrated strong performance in commodity and foreign exchange forecasting, though they are prone to overfitting when trained on noisy intraday data and can exhibit instability in training (Lim et al., 2021). To address these limitations, hybrid architectures that combine LSTM feature extraction with gradient-boosted decision trees, such as XGBoost (Chen and Guestrin, 2016) and CatBoost (Prokhorenkova et al., 2018), have emerged. These models use the temporal learning of LSTMs together with the nonlinear refinement capabilities of boosted ensembles, and studies report improved robustness and predictive accuracy in financial markets (Livieris et al., 2020; Oukhouya et al., 2024).

More recently, Transformer-based architectures have been proposed for time series forecasting. The Informer model Zhou et al. (2021) introduces efficient sparse attention mechanisms designed for long-sequence modelling, whilst PatchTST (Nie et al., 2022) applies patching to improve local temporal con-

text learning. Although these models achieve strong benchmark performance, recent surveys indicate that Transformers do not consistently outperform hybrid or recurrent approaches in intraday financial settings, especially under noisy conditions (Lim and Zohren, 2021).

A recurring issue in the literature is the use of random splits or short back-tests, which can introduce temporal leakage and inflate reported accuracy (Bergmeir and Benítez, 2012). Studies recommend walk-forward and block-wise validation to ensure realistic evaluation under non-stationary market behaviour.

To address the inconsistencies identified in the literature and progress this research direction, this paper evaluates hybrid LSTM-boosted models (LSTM+XGBoost and LSTM+CatBoost) against a Transformer-based Informer model for intraday XAU/USD forecasting across multiple prediction intervals using a leakage-resistant walk-forward validation framework.

3 PROPOSED APPROACHES

3.1 Data Used

This research employs intra-day historical bid and ask price data for XAU/USD, obtained from Dukascopy Bank SA (2025), a reliable source for high-frequency market data. The dataset spans from 1st January 2024 to 29th March 2025, covering over 130,000 tick-derived observations across three different candlestick resolutions (5-minute candles, 15-minute candles, and 1-hour candles).

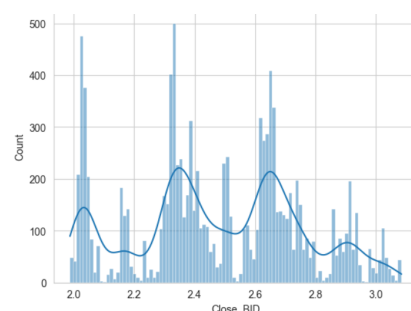


Figure 1: Distribution 1-hour Close-Bid prices.

Each record contains Open, High, Low, Close, and Volume attributes for both bid and ask sides of the market. After alignment and merging, the dataset forms a synchronised bid–ask structure, ensuring each timestamp accurately reflects the market’s microstructure.

The histogram in Figure 1 illustrates the distribution of Close Bid prices in the 1-hour candlestick dataset.

3.2 Data Preprocessing

Data preprocessing in this study was designed specifically for high-frequency bid–ask data, ensuring that the raw tick-derived candlesticks were converted into stable, synchronised and model-ready sequences. This process included merging and cleaning the bid and ask streams, engineering volatility and liquidity related features, applying scale-appropriate normalisation to price and volume series, and optimising memory usage to enable efficient walk-forward training across all timeframes.

3.2.1 Merging and Cleaning

The raw bid and ask datasets were first aligned using their timestamp columns to ensure full temporal synchronisation. Missing or inconsistent rows were dropped to maintain continuous time series integrity. For the 5-minute dataset, after merging, 12,656 rows ($\approx 10\%$ of the merged dataset) were found to contain missing Bid-side values and were removed, resulting in a clean and fully aligned dataset of 118,096 rows. In contrast, both the 15-minute and 1-hour datasets required no row removal after merging, as the Ask and Bid streams were perfectly aligned and contained no missing values.

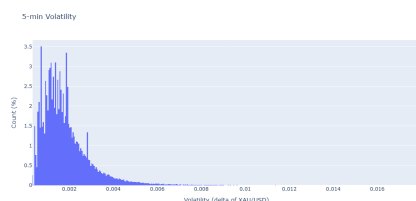


Figure 2: 5-minute candlestick volatility distribution.

This histogram in Figure 2 illustrates the empirical distribution of 5-minute price volatility for XAU/USD, computed as the absolute difference between the high ask and low bid prices within each interval.

3.2.2 Feature Engineering

To enhance predictive power, several engineered features were generated in addition to the standard OHLC and volume metrics:

- Volume Delta = Volume Ask – Volume Bid, capturing buy/sell pressure.

- Absolute Volume Delta: quantifying total liquidity imbalance.
- Price Spread Features: representing the difference between bid and ask OHLC prices, to reflect short-term liquidity and volatility.
- New Day and New Week binary indicators: marking transitions across trading sessions and weekly cycles.

These engineered features enrich the temporal and structural information available to each forecasting model, enabling better recognition of intraday and seasonal market behaviours.

Two target variables were defined for prediction:

- Y_High_Bid: maximum sell opportunity.
- Y_Low_Ask: minimum buy opportunity.

3.2.3 Normalisation

To maintain consistent feature scaling and stabilise model training:

- All price features were divided by 1000, constraining them roughly within the range [1.9, 2.8].
- Volume features were normalised by dividing each value by the maximum training-set volume.
- Delta-based features retained their natural range (0.001–5.981) since they already exhibit small-scale variation.

Figure 3 shows normalised price trends over the recent 1000 records in the 15-minute dataset.



Figure 3: 1000 observations of 15min price trends.

3.2.4 Memory Optimisation

Given the dataset’s size, numerical columns were downcast to 32-bit floats and smaller integer types, cutting total memory usage by half. This optimisation ensured efficient execution across multiple walk-forward iterations and improved computational scalability.

The comparison in Figure 4 shows how volatility widens as the timeframe increases. The 5-minute series exhibits the tightest concentration of small volatility values, while the 15-minute distribution

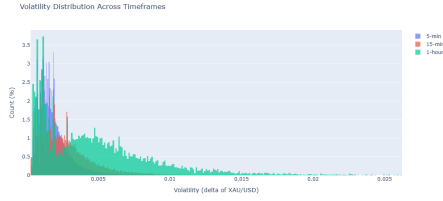


Figure 4: Volatility distribution across all timeframes.

spreads further, and the 1-hour distribution displays a significantly heavier tail.

3.3 Time-Series Validation Setup

To mimic realistic market conditions, walk-forward validation was employed rather than traditional random splits. This evaluation method maintains temporal ordering, ensuring that the model only sees past data during training and is tested on future unseen periods.

Two specialised splitting functions were designed:

- For hybrid models (LSTM–XGBoost, LSTM–CatBoost): The function generates 3D tensors (X_{train} , X_{test} , y_{train} , y_{test}) across multiple rolling blocks.
- For the Informer model: The splitter creates separate CSV files for each sequential block, aligning with the model’s original GitHub implementation (Zhou et al., 2020).

Table 1: Walk-forward block configuration per data frame.

Dataset	Total Length	Train Size	Test Size	Nb of blocks
5-min	118,096	16,500	2,500	7
15-min	43,584	10,000	2,000	4
1-hr	10,896	5,000	800	2

Table 1 summarises the block configurations across datasets and intervals. This strategy prevents data leakage, preserves temporal dependencies, and provides statistically fair model comparisons.

3.4 Model Implementation

Three forecasting models were implemented and compared under identical walk-forward validation protocols, with evaluation performed on High_Bid and Low_Ask targets across 5, 15, 30 and 60-minute intervals.

3.4.1 Hybrid LSTM–XGBoost

This model integrates a Long Short-Term Memory (LSTM) network with Extreme Gradient Boosting (XGBoost) regressors. The LSTM acts as a feature

extractor, transforming temporal sequences into compact 2D feature embeddings. These embeddings are passed to two independent XGBoost regressors, one for High_Bid, another for Low_Ask.

Architecture:

- 1 LSTM layer with 64 units and ReLU activation.
- 1 Dense layer (2 outputs).
- Optimiser: Adam.
- Loss: Mean Squared Error (MSE).
- Regularisation: Early stopping based on validation loss.

After each walk-forward block, RMSE, MAE and R^2 were recorded and averaged to yield stable performance estimates.

3.4.2 Hybrid LSTM–CatBoost

The LSTM–CatBoost hybrid follows an identical workflow, substituting XGBoost with CatBoost, which efficiently handles categorical features and reduces overfitting through ordered boosting.

The LSTM architecture remains the same (64 hidden units + 2-output dense layer). This setup is lightweight but expressive enough to extract temporal signals before passing them to CatBoost regressors. Empirically, this hybrid demonstrated improved stability on non-stationary data and outperformed pure LSTM models.

3.4.3 Informer Transformer

The Informer model represents a modern Transformer-based architecture optimised for long-sequence time series forecasting.

It introduces:

- ProbSparse Attention: to selectively focus on the most informative time steps, reducing quadratic attention cost to near-linear.
- Generative Decoder: that forecasts long horizons efficiently.

The model was adapted directly from the official Informer GitHub repository (Zhou et al., 2020), with configurations tailored to this study’s intervals and feature dimensions.

Like the hybrid models, Informer was trained block-wise in the walk-forward setup, ensuring fair temporal validation.

After training, performance metrics were averaged across all walk-forward blocks to ensure consistency and reduce the effect of individual anomalies.

3.5 Evaluation Metrics

Model performance was evaluated using three key regression metrics across all horizons and datasets:

- Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

Reflects average prediction error magnitude in the same units as the target.

- Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

Penalises larger deviations, suitable for financial data with volatility spikes.

- Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

Measures the proportion of variance explained by the model.

4 RESULTS

This section presents the experimental outcomes for the three forecasting models: the hybrid LSTM-XGBoost, the hybrid LSTM-CatBoost and the Informer architecture. All models were evaluated on XAU/USD bid-ask price data sampled at 5-minute, 15-minute and 1-hour time-frames, with prediction intervals ranging from 5 to 60 minutes ahead.

Performance was assessed using RMSE, MAE and R^2 for both High_BID and Low_ASK targets. All reported results correspond to averages across walk-forward validation blocks, ensuring generalisation across unseen temporal segments.

4.1 Results on the 5-Minute Dataset

The 5-minute dataset contained the highest-frequency information, capturing short-term market fluctuations with over 100,000 data points. At this granularity, the hybrid LSTM-CatBoost model consistently achieved the best performance across all horizons (Table 2).

For the next-5-minute forecast, LSTM-CatBoost reached $RMSE \approx 0.0097$ and $MAE \approx 0.0061$, with $R^2 \approx 0.80$, outperforming both LSTM-XGBoost ($RMSE$

Table 2: LSTM-XGBoost metrics on 5-minute dataset.

Result	Next 5-min	Next 15-min	Next 30-min	Next 60-min
RMSE high bid	0.0124	0.0097	0.0104	0.0122
RMSE low ask	0.0123	0.0099	0.0113	0.0125
MAE high bid	0.0068	0.0058	0.0064	0.0079
MAE low ask	0.0070	0.0061	0.0071	0.0125
R^2 high bid	0.6191	0.8028	0.7793	0.7100
R^2 low ask	0.6303	0.7968	0.7354	0.6938

≈ 0.012) and Informer ($RMSE \approx 0.27$) (Table 3). Although Informer’s R^2 values remained relatively competitive (≈ 0.76 – 0.79), its absolute errors were substantially larger, suggesting it captured general directionality but failed to approximate precise magnitudes.

Table 3: LSTM-CATBoost metrics on 5-minute dataset.

Result	Next 5-min	Next 15-min	Next 30-min	Next 60-min
RMSE high bid	0.0097	0.0095	0.0101	0.0120
RMSE low ask	0.0097	0.0097	0.0105	0.0122
MAE high bid	0.0061	0.0057	0.0062	0.0079
MAE low ask	0.0061	0.0059	0.0066	0.0080
R^2 high bid	0.7992	0.8124	0.7973	0.7157
R^2 low ask	0.7934	0.8088	0.7809	0.7067

As the forecasting interval extended to 15, 30 and 60 minutes, both hybrids maintained stability, with R^2 values remaining above 0.7, while Informer’s performance degraded rapidly ($RMSE > 0.3$, $R^2 \approx 0.5$).

These results indicate that transformer-based architectures like Informer struggle with high-frequency volatility, whereas hybrid recurrent-boosting models retain robustness (Table 4).

Table 4: Informer metrics on 5-minute dataset.

Result	Next 5-min	Next 15-min	Next 30-min	Next 60-min
RMSE high bid	0.2795	0.3196	0.3421	0.3495
RMSE low ask	0.2726	0.3220	0.3346	0.3510
MAE high bid	0.1702	0.2028	0.2255	0.2312
MAE low ask	0.1626	0.2033	0.2162	0.2334
R^2 high bid	0.7637	0.5739	0.5838	0.5202
R^2 low ask	0.7976	0.5598	0.6190	0.4907

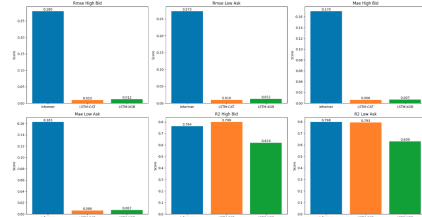


Figure 5: Next 5-minute metrics on 5-minute dataset

Figures 5, 6 and 7 compare the performance of Informer, LSTM-CAT, and LSTM-XGB models for the 5-minute dataset at a 5-minute forecasting horizon. The results show that the hybrid LSTM models—especially, LSTM-CAT, achieved significantly lower RMSE and MAE values and higher R^2 scores

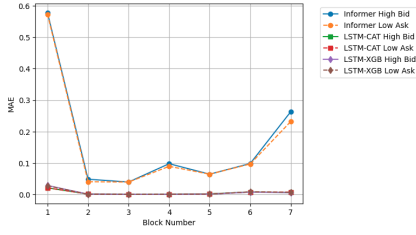


Figure 6: MAE across blocks for 5-minute forecasts

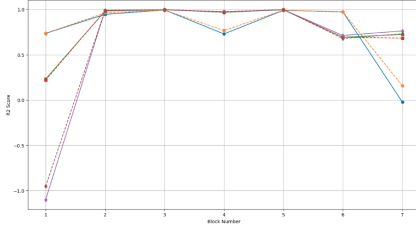


Figure 7: R^2 across blocks for 5-minute forecasts

than Informer, confirming their superior accuracy in short-term bid–ask price prediction.

4.2 Results on the 15-Minute Dataset

The 15-minute dataset ($\approx 43,000$ samples) has balanced granularity and noise reduction, offering a clearer signal-to-noise ratio for learning temporal dependencies.

Table 5: LSTM-CATBoost metrics on 15-minute dataset

Result	Next 15-min	Next 30-min	Next 60-min
RMSE high bid	0.0051	0.0099	0.0071
RMSE low ask	0.0051	0.0100	0.0073
MAE high bid	0.0032	0.0070	0.0048
MAE low ask	0.0031	0.0067	0.0046
R^2 high bid	0.9703	0.8598	0.9484
R^2 low ask	0.9714	0.8523	0.9458

Table 5 shows that LSTM–CatBoost hybrid achieved exceptional results for short-term horizons, producing $\text{RMSE} \approx 0.0051$, $\text{MAE} \approx 0.0032$, and $R^2 \approx 0.97$, effectively explaining nearly all variance in both bid and ask prices.

Table 6: LSTM-XGBoost metrics on 15-minute dataset

Result	Next 15-min	Next 30-min	Next 60-min
RMSE high bid	0.0084	0.0079	0.0070
RMSE low ask	0.0094	0.0078	0.0074
MAE high bid	0.0060	0.0055	0.0048
MAE low ask	0.0065	0.0051	0.0048
R^2 high bid	0.8835	0.9328	0.9510
R^2 low ask	0.8726	0.9337	0.9450

The LSTM–XGBoost also performed strongly ($R^2 \approx 0.88\text{--}0.93$), particularly at longer intervals (30–60 minutes), while Informer again recorded higher errors

($\text{RMSE} > 0.16$) despite acceptable R^2 values (Tables 6 and 7).

Table 7: Informer metrics on 15-minute dataset

Result	Next 15-min	Next 30-min	Next 60-min
RMSE high bid	0.1670	0.2414	0.2132
RMSE low ask	0.1743	0.2367	0.2114
MAE high bid	0.1222	0.1919	0.1628
MAE low ask	0.1280	0.1867	0.1594
R^2 high bid	0.8902	0.6162	0.7181
R^2 low ask	0.8819	0.6557	0.7296

At longer intervals, performance convergence between the two hybrid models was observed. For 60-minute forecasts, both achieved nearly identical metrics ($R^2 \approx 0.94\text{--}0.95$), highlighting the hybrids’ ability to maintain forecasting stability over medium-term horizons.

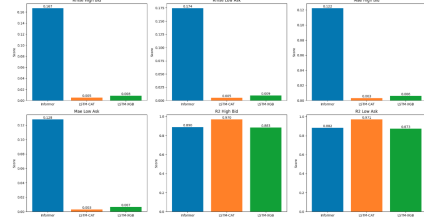


Figure 8: Next 15-minute metrics on 15-minute dataset

Figures 8, 9 and 10 compare Informer, LSTM-CAT, and LSTM-XGB on the 15-minute dataset for a 15-minute forecast horizon. The LSTM-CAT model clearly outperforms the others, achieving the lowest RMSE and MAE values and the highest R^2 scores, demonstrating superior precision and stability in mid-term intraday price prediction.

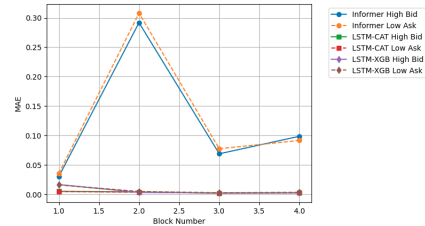


Figure 9: MAE across all blocks for the 15-min forecast using 15-min dataset

4.3 Results on the 1-Hour Dataset

The hourly dataset ($\approx 10,800$ samples) offered a coarse-grained market view, limiting short-term variability capture.

As shown in Table 8 both hybrid models produced moderate accuracy: LSTM–XGBoost achieved $\text{RMSE} \approx 0.035$, $\text{MAE} \approx 0.025$, and $R^2 \approx 0.53$,

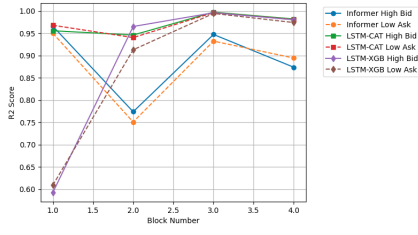


Figure 10: R^2 across all blocks for the 15-minute forecasting (15-minute dataset)

Table 8: LSTM-XGBoost, LSTM-CATBoost, Informer metrics using 1-hour dataset

Result	Next 60-min		
	LSTM-XGBoost	LSTM-CATBoost	Informer
RMSE high bid	0.0352	0.0356	0.3671
RMSE low ask	0.0380	0.0381	0.3398
MAE high bid	0.0242	0.0246	0.2982
MAE low ask	0.0263	0.0261	0.2746
R^2 high bid	0.5365	0.5262	0.3791
R^2 low ask	0.4688	0.4655	0.4683

closely mirrored by LSTM-CatBoost. Informer again performed significantly worse ($\text{RMSE} > 0.33$, $R^2 \approx 0.38$ – 0.46), confirming that it is less effective on lower-frequency financial data with fewer training instances.

Figure 11 illustrates model performance on the 1-hour dataset for the next 60-minute forecast horizon. Both hybrid models (LSTM-CAT and LSTM-XGB) show comparable results with moderate accuracy, while the Informer performs significantly worse.

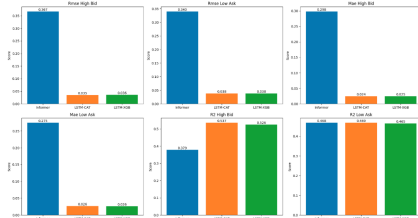


Figure 11: 15-minute forecast metrics (1-hour dataset)

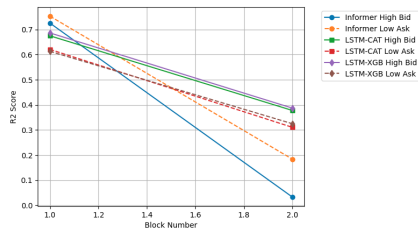


Figure 12: R^2 across all blocks for the 15-min forecast

For validation, the results obtained in this research

were compared with those reported in a state-of-the-art study with similar objectives (Ji, 2025), which reported RMSE values of ≈ 24.0 on the test set and an R^2 of 0.92 when predicting gold prices using a hybrid LSTM-XGBoost model. In contrast, the proposed models in this work achieved higher R^2 scores (up to 0.97 for LSTM-CATBoost) (Figure 12). These findings indicate that, even at finer temporal resolutions and under a walk-forward validation framework, the hybrid boosting-enhanced LSTM architectures deliver superior predictive accuracy compared to previously published approaches.

5 CONCLUSIONS

This study investigated short-term forecasting of XAU/USD bid and ask prices using three advanced architectures: Informer, hybrid LSTM-XGBoost and hybrid LSTM-CatBoost models. Through systematic preprocessing, multi-horizon evaluation, and walk-forward validation across three candlestick resolutions (5-minute, 15-minute and 1-hour), the research demonstrated the comparative strengths and limitations of each approach.

Empirical results revealed that the hybrid LSTM models consistently outperformed the Informer across all datasets, confirming the advantage of combining temporal sequence learning with gradient-boosted regression. The LSTM-CatBoost achieved the most accurate and stable forecasts, particularly on the 15-minute dataset, which offered the optimal balance between volatility capture and noise reduction. These findings highlight the continued relevance of hybrid deep learning-boosting architectures for high-frequency commodity forecasting.

Future work may extend this framework by integrating external market drivers such as news sentiment, macroeconomic indicators, and order book depth, or by experimenting with alternative Transformer-based architectures (e.g., FEDformer, PatchTST). Further exploration of ensemble or multimodal forecasting approaches could also enhance predictive robustness and real-world applicability in financial decision-making.

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